

DRY-RUN GUIDE | lowertriangle.py

Source: Python\3rd_Belt for practice\Matrix\lowertriangle.py

CORE BEHAVIOR

Checks that every element above the main diagonal is zero.

REPRESENTATIVE INPUT

[[1,0],[2,3]].

STEP TRACE

Only above-diagonal cell is [0][1]=0; all checks pass.

EXPECTED OUTPUT / RETURN

'Yes'

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\3rd_Belt for practice\Matrix\lowertriangle.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Check whether a matrix is lower triangular.
002 | # Add your solution here.
003 |
004 | def is_lower_triangular_matrix(matrix):
005 |     m = len(matrix)
006 |     n = len(matrix[0]) if m > 0 else 0
007 |
008 |     for i in range(m):
009 |         for j in range(n):
010 |             if i < j and matrix[i][j] != 0:
011 |                 return False
012 |
013 |     return True
014 |
015 | try:
016 |     m, n = map(int, input().split())
017 |     matrix = [list(map(int, input().split())) for _ in range(m)]
018 |
019 |     print("Yes" if is_lower_triangular_matrix(matrix) else "No")
020 |
021 | except ValueError:
022 |     print("Invalid input. Please follow the input format: '<rows> <cols>' followed by the matrix rows.")
023 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.